

Ghost Identity Hunter — Tooling Inventory, Installation Guide and Provider Gap

Author: Dhayanidhi Rajasekaran **Repository:** <https://github.com/dhayarajas/gih> **Date:** 2026-08-02

1. Scope

Part A lists every tool and service Ghost Identity Hunter (gih) actually invokes, with the installation command for each and the check that proves it is wired up. Part B lists the Maltego data providers gih does **not** use, so the coverage boundary is explicit.

Sources: `docs/T00L_COVERAGE.md`, `docs/EXTERNAL_T00LS.md`, `config/config.yaml`, `requirements*.txt`, `Dockerfile.kali`, and the published Maltego Transform Hub catalogue (<https://www.maltego.com/transform-hub/>).

2. Base installation

Python 3.10+ is required.

```
git clone https://github.com/dhayarajas/gih.git
cd gih
pip install -r requirements.txt
```

Runtime Python dependencies (all installed by the command above): `click`, `networkx`, `phonenumbers`, `requests`, `Brotli`, `beautifulsoup4`, `Pillow`, `pyvis`, `Jinja2`, `tabulate`, `pyyaml`, `numpy`, `urllib3`.

Optional extras, each degrading cleanly when absent:

```
pip install -r requirements-dev.txt      # pytest, pytest-cov, ruff
pip install -r requirements-optional.txt # face matching (dlib, ~3 GB) + Neo4j driver
```

Verify the external tool inventory at any time — this is the authoritative check and prints exactly what the host has:

```
python -m src.cli investigate --check-tools
```

Part A — Tools gih uses

3. Integrated external tools (15)

Each of these has a real output parser in `src/modules/external_tools.py` and is dispatched by `_process_external_tools`. A missing binary is never scheduled; the investigation continues without it.

#	Tool	Used for	Install command	Verify
1	sherlock	username → account presence	<code>pip install sherlock-project</code>	<code>sherlock --version</code>
2	maigret	username → account presence (wider site list)	<code>pip install maigret</code>	<code>maigret --version</code>
3	osrframework (usufy)	username → social account presence	<code>pip install osrframework</code>	<code>usufy --help</code>
4	holehe	email → registered services	<code>pip install holehe</code>	<code>holehe --help</code>
5	theHarvester	domain → emails + subdomains	Kali: <code>sudo apt install theharvester</code> · elsewhere: <code>pipx install git+https://github.com/laramies/theHarvester</code>	<code>theHarvester -h</code>
6	subfinder	domain → subdomains	<code>go install -v github.com/projectdiscovery/subfinder/v2/cmd/subfinder@latest</code>	<code>subfinder -version</code>
7	sublist3r	domain → subdomains	<code>pip install sublist3r</code> (Kali also packages it as <code>sublist3r</code>)	<code>sublist3r -h</code>
8	amass	domain → subdomains (passive)	Kali: <code>sudo apt install amass</code> · elsewhere: <code>go install -v github.com/owasp-amass/amass/v4/..@master</code>	<code>amass -version</code>
9	whois	domain → registrar, dates, registrant	<code>sudo apt install whois</code>	<code>whois example.com</code>
10	dig	domain → A / MX / NS / TXT records	<code>sudo apt install dnsutils</code>	<code>dig example.com +short</code>

#	Tool	Used for	Install command	Verify
11	whatweb	domain → web technologies, resolved IP	<code>sudo apt install whatweb</code>	<code>whatweb --version</code>
12	nmap	IP → open ports and service versions	<code>sudo apt install nmap</code>	<code>nmap --version</code>
13	shodan	IP → host intelligence (<i>API key required</i>)	<code>pip install shodan</code> then <code>shodan init <API_KEY></code>	<code>shodan info</code>
14	exiftool	image → GPS, camera, timestamps	<code>sudo apt install libimage-exiftool-perl</code>	<code>exiftool -ver</code>
15	wayback_machine	domain → archived URLs	none — HTTP call to <code>web.archive.org</code> CDX API	reachable network

Notes worth knowing before relying on a row:

- `osrframework` exposes no binary of its own; gih detects and runs `usufy`. It is restricted to the `social` platform tag with 32 threads (full list far exceeds the time budget; its timeout is 180 s).
- `amass -passive` often exceeds its 60 s budget and returns nothing; `subfinder`, `sublist3r` and `theHarvester` cover the same role.
- `wayback_machine` needs no binary and reports available whenever the network is up, but its CDX query commonly takes 9–31 s against a 30 s timeout, so archived URLs are absent from many runs.
- `nmap` runs `-Pn -F -sV --version-light`, i.e. no `-sS`, so **no root privileges are needed**.
- `shodan` reads the key stored by `shodan init`; no API key is ever placed on a command line.

4. Built-in modules and plugins (no installation)

These ship with the repository and need only the Python dependencies above.

Capability	Module	Notes
Phone validation / carrier / region	<code>src/modules/phone_osint.py</code>	uses <code>phonenumbers</code>
Email OSINT and breach lookup	<code>src/modules/email_osint.py</code> , <code>email_breach_plugin</code>	Have I Been Pwned; key optional
Username enumeration (built-in)	<code>src/modules/username_search.py</code>	independent of <code>sherlock/maigret</code>
Full-name seed expansion	<code>orchestrator + investigation.strict_match</code>	generates username candidates
Google Dorks	<code>src/modules/google_dorks.py</code>	Custom Search API when keyed, scraping otherwise
Image EXIF, hashing, reverse-search links	<code>src/modules/image_search.py</code>	link generation only, no image API

Capability	Module	Notes
Face similarity	<code>src/modules/image_match.py</code>	needs <code>requirements-optional.txt</code> (dlib)
LeakOSINT breach records	<code>src/modules/leakosint.py</code>	paid token required; leads the report when present
Correlation, confidence, risk scoring	<code>src/correlation/linker.py</code>	core, always on
Interactive graph	<code>src/graph/visualizer.py</code>	<code>pyvis</code> / <code>networkx</code>
HTML / JSON / CSV / PDF reports	<code>src/reporting/</code>	4 HTML templates + redaction
Optional graph backend	Neo4j	<code>pip install neo4j</code> , then <code>use_neo4j: true</code>

5. API keys

No key is ever written to a command line, a report, a log, or the database.

Service	Configuration	Behaviour without it
Shodan	<code>plugins.shodan.api_key</code> or <code>SHODAN_API_KEY</code> , plus <code>shodan init</code>	plugin reports unavailable, run continues
LeakOSINT	<code>plugins.leakosint.api_key</code> , or <code>LEAKOSINT_API_TOKEN</code> / <code>LEAKOSINT_API_KEY</code>	skipped, no Breach Records section rendered
Have I Been Pwned	<code>plugins.email_breach.api_key</code>	optional; unauthenticated calls are rate-limited
Google Custom Search	<code>google_api_key</code> + <code>google_cx</code>	falls back to scraping
Etherscan, GeoNames	<code>plugins.*.api_key</code>	declared only — no integration exists , never scheduled

6. Declared but deliberately not invoked (20)

`ToolChecker` knows 36 tools; 15 are integrated and 1 (`leakosint`) runs through the plugin path. The rest are detected and reported by `--check-tools` but never dispatched, each with a reason recorded in `UNIMPLEMENTED_TOOLS`:

`social_analyzer`, `emailharvester`, `recon-ng`, `spiderfoot`, `ghunt`, `photon`, `metagoofil`, `etherscan`, `geonames`, `wappalyzer`, `masscan`, `nikto`, `sqlmap`, `curl`, `wget`, `nslookup`, `tor_browser`, `flagfox`, `user_agent_switcher`, `google_dorks` (handled in-process, not as a subprocess).

7. One-command environment (recommended)

The Kali image installs the whole tool chain, so nothing above has to be installed by hand:

```
docker build -f Dockerfile.kali -t gih:kali .
docker run --rm -it -v "$HOME/.ghost_hunter:/home/ghosthunter/.ghost_hunter" gih:kali
```

Equivalent bare-metal one-liner for a Debian/Ubuntu host:

```
# in the Ubuntu/Debian repositories
sudo apt update && sudo apt install -y nmap whois dnsutils whatweb libimage-exiftool-perl
python3-pip golang-go

# PyPI
pip install sherlock-project maigret holehe sublist3r osrframework shodan

# not packaged for Ubuntu – install from source
pipx install git+https://github.com/laramies/theHarvester
go install -v github.com/projectdiscovery/subfinder/v2/cmd/subfinder@latest
go install -v github.com/owasp-amass/amass/v4/...@master
```

On Ubuntu 22.04 `apt` carries `nmap`, `whois`, `dnsutils`, `whatweb` and `libimage-exiftool-perl` but **not** `amass`, `theharvester` or `sublist3r` — those three are packaged only on Kali, which is why the Docker image is the shorter route.

Part B — Maltego providers gih does not use

Of the 99 providers published on Maltego's Transform Hub, gih reaches four (**Have I Been Pwned**, **Shodan**, **Wayback Machine**, **Dorking**) and has weaker or stub equivalents for three (**Google Social Network Transforms** → `sherlock/maigret`; **TinEye / Image Analyzer** → reverse-search links only; **Etherscan** → declared, no plugin). The remaining 92 have no path in gih today.

Malware / sandbox / threat intelligence

- VirusTotal Public API
- VirusTotal Premium API
- Hybrid Analysis
- Intezer Analyze
- PolySwarm
- Cisco Threat Grid
- CrowdStrike ThreatGraph
- CrowdStrike Intel
- Mandiant
- Abuse.ch URLhaus
- ThreatCrowd
- ThreatMiner
- ThreatConnect
- OpenCTI
- ATT&CK-MISP
- Anomali
- Intel 471
- Recorded Future
- Cybersixgill

Vulnerabilities

- NIST NVD
- Vulners
- Censys
- GreyNoise Enterprise
- GreyNoise Community
- Elemendar
- Team Cymru

Infrastructure, DNS and passive data

- DomainTools Iris Investigate
- DomainTools Enterprise
- Farsight DNSDB
- ZETAnalytics Massive Passive
- WhoisXML API
- Host.io
- IPInfo
- MaxMind
- PeeringDB
- SSL Certificate Transparency Transforms
- DNSTwist
- alphaMountain
- HYAS Insight
- CrowdSec
- Criminal IP
- Spamhaus Intelligence
- urlscan.io
- SOCRadar
- Silobreaker

Breaches and leaks

- SpyCloud Cybercrime Investigations
- Constella Intelligence
- Darkside
- Hades
- Flashpoint
- Digital Shadows
- ZeroFOX
- Cofense Intelligence

Deep and dark web

- DarkOwl
- Vysion
- Web-IQ

- Corvus Intelligence
- Crowlingo
- Tisane Labs

People and personal identifiers

- Pipl
- FullContact
- PeopleMon
- PhoneSearch
- Epieos
- OSINT Industries
- Hunter.io
- Clearbit
- LittleSis
- LoginsoftOSINT
- FlightAware
- Espy
- Social Links Professional
- Social Links CE
- Google Maps Geocoding

Company, corporate and sanctions data

- OpenCorporates
- Orbis – Bureau Van Dijk
- OCCRP Aleph
- OpenSanctions
- EntitiesUA

Cryptocurrency

- Tatum Blockchain Explorer
- Crystal Intelligence

Phishing and reputation

- Scamadviser
- AbuseIPDB
- AlienVault OTX

Image, content and language

- GeoSpy
- DeepL
- Regex Library

Interoperability, case management, long tail

- STIX 2 Utilities

- Polonious
- Discogs
- The Movie Database

Cheapest of these to add

Free or free-tier, so closable without a commercial contract: **crt.sh / Certificate Transparency, IPInfo** (or MaxMind GeoLite2), **NIST NVD, VirusTotal Public API, Hunter.io, AbuseIPDB, Abuse.ch URLhaus, urlscan.io, OpenSanctions, Etherscan** (already declared), **STIX 2 / MISP export**.

Two model changes unlock most of them: `file_hash` and `cve` artifact types, and an `organisation` entity. Without those, several of the free providers have nowhere to attach their results.

Structurally unreachable without commercial contracts: SpyCloud, Constella, DomainTools, Farsight, Recorded Future, Flashpoint, Cybersixgill, DarkOwl, Intel 471, Mandiant, CrowdStrike, Pipl, Social Links, Orbis, HYAS, Silobreaker, Digital Shadows, ZeroFOX.